

$$\beta_i \sim \text{norm}(\beta_i, \sigma/\sqrt{s_x})$$

$$\beta_i = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\sum (x_i - \bar{x})^2$$

$$s_x = \left(\sum (x_i - \bar{x})^2 \right)^{1/2}$$

$$\sigma_i = \left(\frac{s_y}{s_x} \right)^{1/2}$$

$$s_y = \left(\sum (y_i - \beta_0 - \beta_1 x_i)^2 \right)^{1/2}$$

$$t_{n-2} \sim \frac{s_x (\beta_i - \beta_i^*)}{\sigma_i}$$

$$\frac{\beta_i - \beta_i^*}{\sigma_i / \sqrt{s_x}} \sim t_{n-2}$$